

Vector Clay Printer: A Dual-Material 3D Clay Printer with Steerable Inner Nozzle and Concentric Piston

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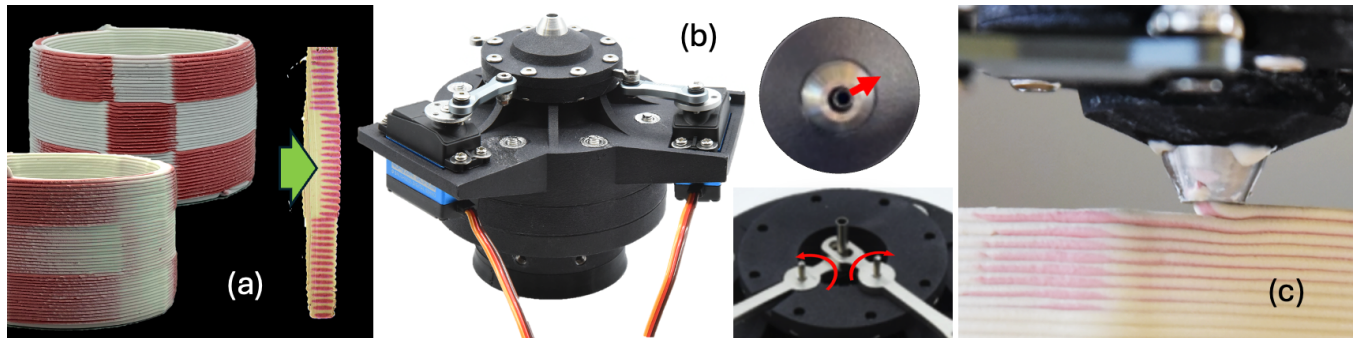


Figure 1: (a) Graded / high contrast textured ceramic print; (b) Servo controlled inner nozzle "vectoring"; (c) Double extrusion

Abstract

We present a dual-extruder clay printer with a fully steerable concentric nozzle. Through precise directional control of the inner needle, the system enables complete encapsulation of one material inside the other, or its controlled exposure on the printed object surface. The system produces graded or high-contrast textured surfaces in a single pass before firing. The movable inner nozzle enables mechanical mixing at the extruder tip, ensuring optimal water distribution in the colloidal suspension while improving both inter-material adhesion and layer-to-layer bonding. We demonstrate two implementations: one that uses separate syringes and plungers, and a more compact version using concentric reservoirs with a single plunger. Through independent material flow control, precise inner extruder positioning, and adjustable retraction within the outer cone, this research opens new possibilities in multi-material clay printing by harnessing the physics of wet, granular matter directly at the extrusion point.

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1 Introduction

In recent years, numerous innovative multi-material extruder configurations have been proposed opening new engineering and aesthetic possibilities for ceramic 3d printing. While co-extrusion using parallel, concentric, and rotating nozzles [Seibold et al. 2019][Jauk et al. 2023a][Larson et al. 2023], to our knowledge, no system implements precise and continuous control of the horizontal position of the nozzle tip. We propose a concentric double nozzle system with a steerable inner extruder needle, inspired by thrust vectoring engines in propulsion systems. During extrusion, the bendable inner nozzle directs the core material to the sides of the composite filament—including back and forth in the print direction. The system can pattern surfaces with sharp contrast or graded colors (see

Fig. 1), while the motion of the head inside the outer nozzle generates variable directional pressure, improving material adhesion and creating irregular surface texture enhancing layer bonding.

2 Design

We modified a commercial 3D ceramic printer (3D PotterBot 10 Pro) by adding one stepper motor to control a second piston and two servo motors to steer the inner nozzle.

2.1 Extruder Head Vectoring

Multi-material Direct Ink Writing 3D printing has been demonstrated using multi-channel nozzles [Lee et al. 2024][Thi Hai et al. 2022] and even concentric nozzles [Wang et al. 2024][Jauk et al. 2023b]. Our system advances this technique by enabling continuous and precise positional control of the inner extrusion path. A dual-servo actuation mechanism positions a 2.5mm inner nozzle within a 5mm outer nozzle (Fig. 1b). The inner nozzle is trapped at the intersection of two pivoting arms—we chose this over linear actuators to eliminate material leaks and reduce friction in the guiding structure. Given the small displacement of the inner extruder, we assumed a linear relationship between servo angle and axial displacement, achieving an angular resolution of at least 0.2 degrees for a circular motion at a speed of 1 turn/sec. This configuration enables three primary operational modes: full encapsulation of the inner material within the outer clay matrix, controlled partial exposure of material ratios on interior/exterior surfaces, and mechanical mixing through nozzle oscillations. The inner extruder can also be positioned vertically (retracted inside the outer nozzle, protruding, or level with it). However, in the current implementation, this vertical positioning can only be adjusted manually.

2.2 Piston-driven dual extrusion

We tested a double cartridge system that enables separate control of material flows. This system helped us determine the optimal material flow rate ratio and prevent excessive deposition of either material. Based on these findings, we developed an innovative concentric dual cartridge that uses a single stepper motor to extrude both materials, creating a more compact system.

2.3 Software

A customized Grasshopper script controls the printer and extruder by generating g-code commands for both the tool-path and inner extruder motion. The script continuously calculates the surface-normal vector pointing outward, while texture values from an input image determine the amplitude of the offset.

3 Results

We used Dehua porcelain (also known as Blanc de Chine) as the printing medium, with a 5% Taiwanese Red Stain additive for the inner material. This clay was chosen for its plasticity and fine texture, allowing smooth extrusion through a 2.5mm nozzle. Fig and Fig demonstrate a "binary" high-contrast pattern and continuous color gradient, respectively. The cross-section reveals that the red clay (which appears flattened) is fully encased by white clay and can be moved perpendicular to the contour to expose color on either the outside or inside of the print. Moving the print head back

and forth along the printing direction affects the upper and lower thickness of the containing white clay—an effect not yet explored in this work but potentially valuable for controlling adhesion and other mechanical properties.

4 Applications and Future Works

We have demonstrated a modified ceramic 3D printer that uses a mobile, concentric extruder and concentric pistons to pattern surfaces before firing with clean and repeatable results also enabling innovative applications beyond surface printing. Strength of the print or thermal properties can be enhanced by a judicious combination of materials as in [Teng et al. 2024] and by using clays with different shrinkage rates, we could indirectly control forces during firing. An inner clay with high translucency can work by distributing light inside the material in interesting ways, offering new possibilities for expression. Finally, water-soluble or burn-away materials [Yang et al. 2017][Nakagawa et al. 2017][Jauk et al. 2023a] can enable the fabrication of hollow tubular structures as demonstrated in [Jauk et al. 2023a], but with the additional possibility of creating openings at precise locations in the surface. These structures could facilitate pumping water or air for heating or cooling liquids in containers. Using conductive materials create intriguing possibilities such as the fabrication of ceramic antennas [Chen et al. 2022] in one go, or rendering surfaces touch-sensitive through capacitive sensing [Ames et al. 2024].

Future works include full automation of the inner nozzle retraction as well as inclusion of other elements in the end effector, such as high frequency vibration actuators to control the viscosity of the material at the tip. These advancements aim to establish a new class of physics-aware ceramic additive manufacturing systems that actively modulate material properties during deposition.

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